SHAHID SYED ALI

MECHATRONICS ENGINEERING STUDENT

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SUMMARY OF QUALIFICATIONS

- Third-year BEng + MEng (Hons) Mechatronics Engineering student at the University of Nottingham with a 75% First Class Honours result in Second Year.
- Hands-on robotics experience spanning autonomous mobile robots, ROS/Autoware, LiDAR-based perception, mapping, sensor integration, embedded control and robotic arms.
- Practical experience with TARS autonomous delivery robots and EMERGE autonomous buggies, including testing, troubleshooting, autonomous navigation and charging/parking alignment.
- Strong engineering foundation across CAD, electronics, microcontrollers, manufacturing and control systems, with experience taking systems from design and simulation through fabrication and testing.

PROFESSIONAL EXPERIENCE

Robotics Engineering Intern — Jun 2026 – Sep 2026

HelloWorld Robotics Sdn. Bhd. — Kuala Lumpur, Malaysia

- Worked hands-on with TARS autonomous delivery robots and EMERGE autonomous buggies, supporting robot bring-up, operation, testing and troubleshooting.
- Worked with ROS/Autoware workflows, Ubuntu/SSH tools and Lanelet maps; created and tested a Level 2 company-building map for autonomous navigation.
- Tested five EMERGE buggies and tuned autonomous parking goal coordinates to improve charging-receiver and ground-transmitter alignment, enabling autonomous driving without manual takeover.
- Supported TARS solar-charging integration through wiring, BMS and software testing, and assisted robot demonstrations and technical visits.

SELECTED ENGINEERING PROJECTS

Industrial Conveyor Test Rig + Robot Arm Pick & Place — Sep 2025 – Apr 2026

University of Nottingham — Semenyih, Malaysia

- Implemented an Arduino Mega controller for an ICT3/Bytronic peg-ring verification rig, interfacing 24 V industrial devices with 5 V logic through opto-isolated inputs and relay-driven outputs.
- Integrated a Raspberry Pi and robot arm for pick-and-place using forward/inverse kinematics and motion sequencing; designed in SolidWorks and supported simulation and 3D-printed fabrication.

Centrifugal Clutch Design & Manufacture — Sep 2025 – Apr 2026

University of Nottingham — Semenyih, Malaysia

- Modelled parts and full assembly in SolidWorks; produced BS 8888 manufacturing drawings, BOM/cutting list and process sheets.
- Verified centrifugal/spring forces, pressure and torque against a ≥0.3 N·m at 2000 rpm requirement and supported CNC fabrication, fitting, assembly and testing.

Advanced Line-Following Robot Car — Sep 2024 – Dec 2024

University of Nottingham — Semenyih, Malaysia

- Built a 4WD Arduino robot with custom IR sensing, ultrasonic obstacle detection, rotary-encoder distance measurement and MPU6050 orientation/ramp sensing.
- Developed embedded control and sensor-integration code, added LCD live data and HC-05 Bluetooth mobile operation.

EDUCATION

BEng + MEng (Hons), Mechatronics Engineering — University of Nottingham, Malaysia Semenyih, Malaysia Sep 2024 – May 2028

- Third Year | Current
- Second Year — 75% | First Class Honours
- First Year — 73% | First Class Honours

Higher Secondary — Al Yasmin International School, Riyadh, Saudi Arabia 2024 • 90.2% (CBSE-India)

TECHNICAL SKILLS

Robotics & Software: ROS/Autoware, Python, C, NumPy, Ubuntu/Linux, embedded control, autonomous navigation

Perception & Mapping: LiDAR, sensor fusion, 2D mapping, Lanelet maps, rotary encoders, ultrasonic/IMU sensing

Engineering & Hardware: SolidWorks, MATLAB, LTspice, Arduino, Raspberry Pi, robot-arm kinematics, ladder logic, CAD, prototyping, CNC/workshop fabrication

REFERENCES

1. Kennedy Wai • kennedywai@helloworldrobotics.xyz
2. Dr Marwan Nafea • Marwan.Nafea@nottingham.edu.my
3. Ir. Ts. Nandan Ganesh Jayabalan • nandangan@nottingham.edu.my